Chandana Vishnu Jayanth

Hyderabad, India • jayanthvishnu9@gmail.com • linkedin.com/in/vishnujayanth • github.com/vishnuatgit

Summary

B.Tech CSE (AI/ML) student applying computer science fundamentals to build production async distributed systems, ML data pipelines, and AI-integrated backends. Interned at VSC DigiTech shipping a scalable 5-module ML platform; built DeepBell, an async SRE telemetry system with LangChain-powered automated RCA.

Education

Sri Sri University, Cuttack, Odisha CGPA: 7.7 / 10.0

B.Tech, Computer Science and Engineering (Artificial Intelligence and Machine Learning)

Penultimate Year Student | Expected Graduation: June 2028

Coursework: Data Structures & Algorithms, Database Management Systems, Computer Networks & Socket Programming, Machine Learning, Operating Systems (Concurrency & Synchronization), Software Design.

Work Experience

Machine Learning Development Intern February 2026 – March 2026

VSC DigiTech Pvt. Ltd., Hyderabad, India

- Led collaboration across cross-functional engineering teams to architect and ship a scalable, 5-module production ML platform processing 36,893 transaction records; deployed under NDA.
- Engineered a 3-dimensional RFM ETL feature pipeline in Pandas/NumPy, resolved raw schema inconsistencies, and trained a KMeans clustering model to perform data mining and segment customers into cohorts.
- Developed a RandomForest regression model on lag-1/2/3 time-series features to forecast monthly revenue, and integrated a recommendation engine to drive upsell strategies.

Projects

DeepBell — *Async Distributed SRE Telemetry & AI-Integrated Cockpit Dashboard (Complex System)*

github.com/vishnuatgit/DeepBell *FastAPI, Python, LangChain, SQLite, Chart.js, Docker, pytest, Git*

- Integrated LangChain and Google Gemini API to build an AI-integrated automated Root Cause Analysis (RCA) engine, parsing anomalous telemetry streams to generate structured incident reports.
- Engineered a distributed async FastAPI backend with event loops, concurrency controls, multi-threading, and resource synchronization locks, designed for scalability under concurrent load.
- Designed a dark-mode cockpit dashboard using Chart.js to analyze system performance and reliability metrics — uptime, latency, and time-series trends — for real-time debugging and observability.
- Containerized the full system with Docker and automated test validation via GitHub Actions CI/CD pipeline, ensuring reliable deployment across environments.
- Utilized raw HTTP networking sockets and SQLite schema design to track incident states, reducing MTTD by ∼40% under high-concurrency load testing.

SupplyTrace — *Machine Learning Data Pipeline & Orchestrator*

github.com/vishnuatgit/SupplyTrace *Python, Scikit-learn, Pandas, NumPy, pytest, Git*

- Designed and implemented a scalable, modular pipeline architecture in Python — ingesting raw XML, validating schemas, and compiling standardized CSV feature matrices — built to support extensible ML workloads.
- Refactored core ingestion logic into an object-oriented Orchestrator class with clean separation of concerns, applying software design principles for maintainability and testability.
- Integrated a RandomForest classification model using scikit-learn to analyze system feature matrices and perform data analysis to predict supply chain operational risks.
- Wrote automated unit and integration testing suites in pytest covering data parsing, validation rules, and model predictions, achieving consistent end-to-end pipeline reliability.

ArchLens — *Engineering Health Scorer for GitHub Repositories*

github.com/vishnuatgit/ArchLens *FastAPI, SQLAlchemy, Alembic, httpx, PostgreSQL, Docker, Locust*

- Scores public GitHub repos 0–100 across seven engineering dimensions and checks dependencies against OSV.dev.
- Runs as a web app, REST API, CLI and GitHub Action; fixed 30 s load-test stalls (p95 510 ms); 332 tests in CI.

Technical Skills

Languages: Python, Go, Java (Basic), JavaScript (ES6+), SQL, C

Concurrency & Systems: Async programming (async/await), event loops, multi-threading, synchronization locks, distributed systems, networking, Socket Programming, process management, Linux CLI

Software Design & DSA: OOP design patterns, API design, database schema design, Big-O Analysis | Trees, Graphs, Hash Maps, Doubly Linked Lists, LRU Cache, Prefix Trie, Priority Queues *(from scratch in Python)*

Machine Learning & AI: Scikit-learn, RandomForest, KMeans Clustering, Feature Engineering, ETL Pipelines, Pandas, NumPy, LangChain, LLM orchestration

Engineering & Tools: FastAPI, SQLite, Docker, Docker Compose, GitHub Actions, CI/CD, pytest, Git